

GOOGLE CYBERSECURITY CERTIFICATE

Vulnerability Assessment

Google Cybersecurity Professional Certificate · Study Guide

A NIST SP 800-30 Rev. 1 vulnerability assessment of a MySQL database server holding customer, campaign, and analytics data — scoring threat sources by likelihood and severity and prioritizing remediation.

Track	Google Cybersecurity Professional Certificate
Course	Assets, Threats, and Vulnerabilities
Framework	NIST SP 800-30 Rev. 1
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1 System Description & Scope

Component	Details
Hardware	High-performance CPU, 128GB RAM
OS	Latest version of Linux
Database	MySQL
Network	IPv4, connected to other servers on the network
Security	SSL/TLS encrypted connections

Scope: this assessment covers the current access controls of the database server. The server stores customer, campaign, and analytic data used for marketing operations — making it a critical asset requiring strong protection.

2 Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk score
Hacker	Obtain sensitive information via exfiltration	3	3	9 — High
Employee	Disrupt mission-critical operations	2	3	6 — Medium
Customer	Alter or delete critical information	1	3	3 — Low

NOTE Risk Score = Likelihood x Severity (each scored 1-3). Threat sources and events were identified by analyzing the server's open access permissions; severity was weighed against the operational impact of each potential incident.

Remediation strategy

Control	Description
Authentication & Authorization	Strong passwords, RBAC, and MFA to limit user privileges
Encryption in transit	Replace SSL with TLS for all data in motion
IP allow-listing	Restrict database access to corporate office IP ranges
Auditing	Enable audit logging to track all access and modifications

3 Quick Reference



Vulnerability assessment — quick reference

NIST SP 800-30 Rev. 1 risk score = Likelihood (1-3) x Severity (1-3)

1-3 Low 3-4 Low/Medium boundary

6 Medium

9 High

Top hacker risk here: 9 (High) -> prioritize MFA + RBAC + TLS + IP allow-listing